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CHAPTER I

INTRODUCTION

1.1 Background of the Study

In recent years, the global transportation industry shifted significantly toward flexible, on-demand vehicle rentals as an affordable alternative to the heavy financial burdens of traditional car ownership [1, 2]. Driven by rapid advances in internet technology, modern consumer expectations evolved toward fast, self-service digital platforms providing real-time inventory visibility [3, 4]. In developing regions and tourist hubs, this demand sparked rapid growth among small and medium-sized car rental operators, as well as individual vehicle owners participating in car-sharing models to support regional commerce and local mobility [2, 5].

Despite this growth, many local rental businesses faced a significant technology gap, struggling to keep up with modern digital standards due to the high cost and complexity of enterprise software [6]. Instead, they continued to rely on manual operational workflows—such as paper logbooks, whiteboards, spreadsheets, and fragmented instant messaging chats—to manage reservations, vehicle schedules, and transactions [7]. These manual methods resulted in severe operational bottlenecks, including frequent double-booking errors, time-consuming availability checks, slow check-in/check-out procedures, unmonitored maintenance schedules, and disorganized financial recordkeeping [4, 7–10].

1.2 Statement of the Problem

Existing local car rental operations rely heavily on manual tracking tools and fragmented communication channels. This traditional approach creates critical operational bottlenecks and growth constraints for independent fleet owners:

1. **Scheduling Overlaps and Double-Booking:** Reliance on paper logbooks, wall calendars, and unorganized messaging threads introduces data synchronization delays, leading to overlapping vehicle reservations, lost bookings, and damaged business reputations.
2. **Insecure Data Handling and Privacy Risks:** Collecting sensitive customer driver's licenses and payment receipts through personal messaging apps or physical binders creates severe data privacy vulnerabilities, exposing customer records to loss, theft, and regulatory non-compliance.
3. **Fleet Tracking Deficiencies and Billing Errors:** Manual vehicle inspections and hand-calculated rental fees result in frequent pricing errors, uncollected late charges, and missed maintenance schedules that cause unexpected mechanical breakdowns.
4. **Financial Opacity and Scalability Limits:** The absence of a centralized database with real-time reporting leaves business owners without clear visibility into net profits or vehicle ROI, causing paper-based workflows to completely collapse during peak travel seasons.

Without a centralized, web-based management and analytics platform like **InstaCar**, local car rental operators remain trapped in time-consuming administrative routines that limit business growth and lower service reliability.

1.3 Objectives of the Study

1.3.1 General Objective

The primary objective of this study was to design, develop, and evaluate **InstaCar**, a responsive web-based car rental management and analytics system that replaced error-prone manual workflows with an automated digital platform for local vehicle operators. The system resolved critical operational bottlenecks by providing real-time schedule synchronization to eliminate double-booking, automated billing for precise financial tracking, and an interactive analytics dashboard for data-driven fleet decisions. Furthermore, InstaCar enhanced operational security and business transparency through protected file submissions, automated profit-sharing algorithms, and a verified customer review module.

1.3.2 Specific Objectives

Specifically, the study aimed to:

1. Build a Real-Time Car Tracking System: Create a central database tool that automatically tracks whether each car is free, rented, reserved, or in repair, instantly updating schedules to stop double-booking mistakes and keep cars busy.
2. Design an Easy Mobile Website: Build a simple, easy-to-use website that works on phones, tablets, and computers, allowing customers to book cars quickly while giving staff an easy dashboard to manage daily requests.
3. Automate Price Math and Cancellations: Write computer code that automatically calculates rental totals, late fees, extension costs, and cancellation refunds to remove human math mistakes and speed up checkout times.
4. Create an Interactive Color-Coded Calendar: Design an online visual calendar connected directly to live bookings so staff can clearly see active trips, upcoming rentals, and set aside time to wash and repair cars.
5. Build an Automatic Profit-Sharing System for Car Partners: Create an accounting tool that automatically calculates profit splits, commission rates, and repair costs for third-party car owners, providing clear payout reports to build trust.
6. Provide a Central Digital Storage for Customer IDs: Build a safe file upload page that connects driver's licenses, government IDs, and payment screenshots directly to customer profiles, replacing messy messaging apps and paper folders.
7. Create Visual Business Charts and Forecasting Tools: Build an executive analytics dashboard that turns daily sales numbers into clear visual charts, showing popular car models, busy rental seasons, and future market demand to guide smart vehicle purchases.
8. Build a Customer Profile and Review System: Create customer account records that track rental histories, late returns, and real star ratings to lower client hesitation, build company trust, and reward loyal renters.
9. Keep Data Safe and Ensure System Growth: Protect user records using strict password security and access controls while testing the website for speed, safety, and reliability to ensure the software stays fast as the business adds more cars.

1.4 Scope and Limitations

Scope:

- **Core System Setup:** Development of **InstaCar**, a web-based car rental management and analytics system that replaces paper logbooks, physical file cabinets, handwritten bills, and spreadsheets with a centralized digital platform.
- **User Roles & Access:** Implements dedicated access panels for System Administrators (full system control), Third-Party Vehicle Owners (partner dashboard for earnings and commission tracking), and Users (public website for browsing, booking, and document submission).
- **Automated Financial Engine:** Executes server-side price calculations for daily or 10-hour rates, late return penalties, fuel shortfall charges, promo discounts, self-service cancellation refunds, and partner profit-sharing splits with maintenance deductions.
- **Document Safety & Verification:** Provides a secure file upload tool for driver's licenses, government IDs, and payment receipts, featuring a side-by-side staff verification screen, strict file access controls, and scrambled password storage.
- **Analytics Dashboard:** Integrates visual graphs for gross sales, vehicle usage percentages, and car popularity trends.

- **Limitations:**

  - **Physical Hardware & Vehicle Tracking**: The platform operates strictly as a web application and does not connect to onboard GPS boxes, remote engine cutoffs, keyless door locks, automatic key dispensers, or diagnostic sensors for fuel levels, tire pressure, or body damage.
  - **Live Telematics & External Fees**: Excludes live map satellite tracking, real-time speed alerts, automated highway toll collection, parking ticket integration, and physical workshop/mechanic tool inventory tracking.

1.5 Definition of Terms

- **Analytics Dashboard:** An interactive visual interface within the system that synthesizes raw operational and financial data into charts, graphs, and performance indicators to support data-driven decision-making.
- **Automated Billing Engine**: The backend server-side logic that automatically computes base rental rates, hourly extension fees, late return penalties, fuel shortfall charges, promo discounts, and cancellation refunds without manual intervention.
- **Demand Forecasting:** An algorithmic analysis tool that uses past seasonal booking rates and trends to project future vehicle demand during upcoming holidays and quiet business weeks.
- **Double-Booking:** An operational error where the same vehicle is concurrently reserved for two different clients over overlapping date or time ranges.
- **Partner Profit-Sharing (Tie-Up Engine):** An automated accounting feature that calculates net rental distributions, splits commission earnings, and subtracts maintenance deductions between agency administrators and third-party vehicle owners.
- **Responsive Web Architecture:** A web development design methodology ensuring that system interfaces fluidly adjust layouts, forms, and data tables across desktop monitors, tablets, and mobile phone viewports.
- **Role-Based Access Control (RBAC):** A system access mechanism that restricts available pages, functions, and database views according to a user's assigned role—categorized into System Administrators, Third-Party Vehicle Owners, and Renters/Users.
- **Self-Service Cancellation:** A portal feature allowing renters to cancel upcoming reservations directly through their account according to system rules, automatically changing vehicle status back to available in public search results.

- **Vehicle Usage Rate:** A calculated metric expressing the percentage of time a specific car is actively rented out compared to its total available calendar days within a given date range.

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CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Theoretical Foundations of Entropy-Based Anomaly Detection

2.1 Introduction

This chapter reviews related literature and studies that ground the design and development of **InstaCar**, the web-based car rental management and analytics platform proposed for local vehicle operators. Citations follow the IEEE numbered style, in which sources are numbered in the order they are first cited in the text and are listed in the same order in the reference list. The chapter is organized into four bodies of work — foreign literature, foreign studies, local literature, and local studies — followed by a synthesis that identifies the specific gap this study addresses, and a theoretical framework that anchors the system's design and evaluation in established models of technology acceptance, information systems success, and software development life cycles.

2.2 Foreign Studies

- The study by Jyothi and Sheethal (2025) explored the structural development and implementation of an advanced Car Rental Management System engineered to streamline endtoend vehicle rental workflows through total process automation, focusing on constructing an integrated software platform capable of supporting both client-facing utilities—such as real-time vehicle browsing, preference filtering, and secure user authentication—and administrative backoffice functions, including fleet utilization optimization, dispatch scheduling, and automated billing workflows. Complementing this work, recent international research on webbased rental systems (Gour et al., 2025) investigated the design and deployment of automated booking platforms utilizing server-side PHP and relational MySQL architectures to eliminate paperbased logs, proving that serverside request processing and structured relational databases prevent scheduling overlaps, lower administrative processing times, and drastically reduce human error across daily operations. Furthermore, architectural studies on transport Management Information Systems (MIS) emphasize that traditional, non-digitized rental operations suffer from severe operational bottlenecks due to rigid scheduling, untracked payment methods, and zero visibility into fleet utilization, proposing centralized digital frameworks that incorporate automated billing logic, interactive availability grids, and datadriven reporting analytics to transform raw rental records into actionable Key Performance Indicators (KPIs) for strategic fleet expansion.
  Additional global studies on digital document management and Role-Based Access Control (RBAC) in automotive software highlight that providing secure digital upload channels for client verification files—such as government-issued identity cards and proof-of-billing statements—elevates corporate professionalism, speeds up customer onboarding, and protects data integrity compared to physical check-in desks.
  Collectively, these foreign studies directly validate the architectural and functional design of **InstaCar**, as they provide strong empirical and theoretical support for InstaCar’s choice of a

webbased PHP/MySQL tech stack, its server-side 10-hour/24-hour pricing engine, its visual dispatch calendar, its secure document verification queue, and its Analytics Dashboard, reinforcing the core project objective of replacing manual, error-prone tracking with an integrated, highperformance digital alternative.

- The study conducted by Azlan (2020) at Universiti Teknologi MARA explored the inherent disadvantages of manual, paper-based car rental operations, identifying them as timeconsuming, inefficient, and highly vulnerable to physical data loss, misplaced documentation, and cumbersome record retrieval. To resolve these operational bottlenecks, Azlan engineered a webbased Car Rental Service System designed to centralize and digitize reservation workflows using a core technology stack of PHP, HTML, CSS, and a relational MySQL database engine. Through empirical usability testing and post-task respondent questionnaires, the study proved that transitioning from physical record-keeping to a centralized web platform drastically improves operational efficiency, lowers administrative costs, and guarantees long-term data security. Azlan's research is directly relevant to **InstaCar** as it provides a foundational theoretical and technical justification for digitizing vehicle rental management to prevent data loss, eliminate paper trail errors, and streamline customer bookings. Both systems share an identical technical foundation—relying on PHP and MySQL to process complex relational database queries, manage inventory states, and handle user authentication sessions. Furthermore, Azlan's focus on the severe inefficiencies of manual paperwork directly mirrors the core problem statement of the current study regarding local car rental operations in the Philippines. Finally, the usability evaluation methodology demonstrated in Azlan's study serves as a direct procedural model for validating InstaCar's user experience (UX), system usability, and operational efficiency through structured respondent feedback.

- The research by Ogbiti and Aaron (2024) presents the design and rigorous verification of a webbased car rental management system aimed at optimizing operations for both clients and administrators. Developed using a professional stack of PHP, JavaScript, and Bootstrap, the system ensures a high level of responsiveness and dynamic user interaction. A core component of the research is the focus on software reliability, using Microsoft SQL Server and the MSUnit framework to validate critical system features. The authors successfully executed various test cases for user registration, reservation management, and role-based access controls, achieving a perfect 100% pass rate. To ensure high standards of quality, the study utilized a "test traceability matrix" to map specific business requirements to their corresponding technical tests. Additionally, the researchers produced a comprehensive user guide to reduce the learning curve and improve the overall end-user experience. Ultimately, the study showcases how modern technologies and structured testing methodologies can be combined to overcome common software implementation challenges. This study is highly relevant to InstaCar as it validates the use of a similar technical stack, specifically PHP and Bootstrap, to build a professional-grade management platform. The researchers' emphasis on Role Assignment directly mirrors InstaCar's multi-user structure, which distinguishes between the permissions of the System Administrator and the Car Operators. Furthermore, the use of a test traceability matrix provides a methodological blueprint for how the current project can verify its unique 10h/24h pricing logic and commission calculations. By achieving a 100% pass rate in their testing, Ogbiti and Aaron provide an academic benchmark for the reliability and robustness that InstaCar aims to achieve for local car rental firms. Finally, their focus on documentation through user guides reinforces the current study's objective of creating a user-friendly tool that requires minimal technical training.

2.3 Local Studies

- The research paper authored by Golo and Encarnacion (2024) addresses the critical operational inefficiencies plaguing both visiting tourists and local transport providers within the vehicle rental sector of Siargao Island, Philippines. Despite the island's booming tourism industry, the local rental market remains constrained by outdated, manual coordination processes that frequently result in unreliable service delivery, severe booking difficulties, and fragmented fleet oversight. To bridge the gap between these traditional practices and modern technological capabilities, the authors propose the development of an online vehicle rental management system engineered to streamline day-to-day operations and elevate the overall customer experience. The researchers deployed a rigorous mixed-methods approach, combining qualitative data from ethnographic studies and personal interviews with quantitative survey metrics gathered from tourists, residents, and rental providers. This comprehensive methodology allowed the study to thoroughly diagnose the distinct systemic bottlenecks and user demands thriving within the island's unique rental ecosystem. Ultimately, the findings exposed significant inadequacies in manual tracking methods, which in turn illuminated an overwhelming market demand for an automated, accessible, and user-friendly online booking platform. By implementing this advanced software framework, the system automates reservations, optimizes asset tracking, and distributes real-time notifications to drastically reduce human error while maximizing service reliability across the region.

  By deploying this tailored digital infrastructure, the proposed system seeks to actively empower local, independent rental businesses by equipping them with the professional tools necessary to manage their fleets and remain highly competitive in a rapidly accelerating digital age. The research underscores the profound potential of this digital platform to completely revolutionize transportation logistics in Siargao by dramatically enhancing service accessibility, transactional convenience, and consumer satisfaction. Furthermore, transitioning from chaotic manual paper trails to a centralized digital ecosystem allows small-scale operators to eliminate scheduling conflicts and capture a wider demographic of tech savvy travellers. The authors emphasize that the future stability of Siargao's tourism-driven economy relies heavily on the integration of these seamless digital touchpoints to replace high-friction legacy workflows. Consequently, the long-term execution of this web-based management platform is expected to contribute pivotally to the sustainable infrastructure development of the island's hospitality sector. By fostering a more reliable and organized transit framework, the software yields widespread socioeconomic benefits that positively impact international travellers, native business owners, and the broader community alike. Ultimately, Golo and Encarnacion establish that embracing comprehensive technological modernization is no longer an operational luxury, but a core requirement for safeguarding the economic future of top-tier Philippine tourism destinations.

- In 2023, De Jesus conducted an in-depth examination of the car rental industry in the Philippines, specifically focusing on the operations and growth of "Drive Manila." The research aimed to analyze evolving customer behaviors, successful marketing strategies, and current industry trends within the national landscape. A primary conclusion of the study was

that a robust online presence is no longer optional but essential for survival in a competitive market. The author noted that digital interaction acts as a critical driver for enhancing customer engagement and maintaining a competitive edge over traditional agencies. The research highlights a significant shift in the Philippine market where customers have become increasingly reliant on online tools to ensure transaction convenience. By studying Drive Manila's model, the researcher identified that transparency in booking and ease of communication are top priorities for local renters. The study also explored how digital platforms allow businesses to scale faster by reaching a broader demographic across different regions. It emphasizes that the future of the Philippine rental sector lies in the integration of seamless digital touchpoints throughout the customer journey. Ultimately, De Jesus concluded that companies failing to prioritize digital transformation risk losing their market share to more tech-integrated competitors. The study serves as a strategic call to action for local rental providers to adopt comprehensive web-based solutions. The research by De Jesus (2023) and InstaCar are perfectly aligned in their focus on the "Digital Transformation" of the Philippine vehicle rental sector. While De Jesus provides the theoretical and market-based evidence for why digital interaction is essential, InstaCar serves as the practical technical solution to fulfil those specific needs. The study identifies "online presence" as a competitive necessity, which InstaCar achieves by providing a centralized web portal accessible to any user in the country. InstaCar directly implements the "digital interaction" recommended by De Jesus through its integrated features like secure booking and document upload. Unlike the specialized focus of Drive Manila, InstaCar offers a more inclusive "Tie-up" model that allows smaller, independent owners to achieve the same professional digital presence described in the study. While the 2023 study highlights customer reliance on online tools for convenience, InstaCar optimizes this convenience by offering flexible 10-hour and 24-hour rental tiers. Both the study and the project emphasize the importance of customer feedback; De Jesus views it as a marketing strategy, while InstaCar implements it as a core functional feature for fleet quality control. The security concerns mentioned in the study are addressed in InstaCar through a structured identity verification process during the booking phase. InstaCar essentially bridges the gap identified by De Jesus by providing a "hassle-free" interface that eliminates the need for physical office visits. Both works conclude that the future of the local industry is tied to how well a system can manage the digital relationship between the provider and the client. Ultimately, InstaCar takes the strategic insights from the De Jesus study and transforms them into a functional, data-driven software architecture.

- Michael Albino’s 2021 study, "Development of Car Rental Management System with Scheduling Algorithm," focused on automating the core activities of a vehicle rental business to improve reliability. The research addressed the common problems of manual scheduling and inventory tracking by introducing a specialized algorithm. This system was designed to ensure that car transactions were processed without conflicts, making the business operations easier to manage. The researcher adopted the Extreme Programming (XP) Methodology, emphasizing continuous planning and testing throughout the development lifecycle. Key features of the system included modules for transaction management, scheduling, and vehicle inventory control. Upon completion, the application was evaluated by respondents based on specific criteria like speed and the quality of the Graphical User Interface (GUI). The results showed a highly positive response, indicating that users were satisfied with the software’s performance and efficiency. It was specifically noted that the scheduling algorithm performed its intended operations accurately under various test scenarios. The study concluded that the integration of an automated algorithm provided an appropriate solution to the errors found in manual systems. Ultimately, the work proved that a localized digital solution could significantly

optimize the fleet management process for rental businesses. The study by Albino (2021) and InstaCar both prioritize the use of backend logic to solve the complexities of vehicle scheduling in the Philippine market. While Albino utilized a general scheduling algorithm for basic bookings, InstaCar refines this by implementing specific logic for 10-hour and 24-hour rental durations. Both projects emphasize the importance of a clean Graphical User Interface (GUI) to ensure that local users can navigate the system with minimal training. However, InstaCar expands on Albino's model by introducing a "Tie-up" module, allowing multiple independent vehicle owners to list their cars on a single platform. While Albino's system was designed as a centralized management tool for a single company, InstaCar functions as a decentralized marketplace for various providers. Both systems rely on a robust database to track inventory in real-time, though InstaCar adds automated communication gateways for booking notifications. The findings in the 2021 study regarding "Speed" as a key satisfaction metric serve as a direct performance benchmark for the InstaCar backend. Both projects aim to eliminate manual paperwork and human error in the rental process to build better customer trust. Ultimately, InstaCar builds upon the foundation of reliable scheduling by adding layers of user verification and flexible duration tiers that were not present in the 2021 system.

- The 2023 study by Abat, Buen, and Giron, titled "Online Car Rental System," aimed to modernize vehicle booking for travel and business purposes in the local Philippine context of La Union. The researchers focused on creating a Hassle-free platform that could accommodate various travel agendas while being easy for the general public to navigate. A key component of the research was determining the socioeconomic profile of potential users, which helped tailor the system to the community's financial needs. The study utilized a quantitative approach, gathering data from fifty respondents and analyzing results through T-Test and ANOVA statistical tools. Demographic findings revealed that the majority of users were young adults under 25, predominantly from a lower-income bracket earning ₱10,000 or less monthly. Technically, the system underwent a vulnerability test that identified and mitigated five medium and eleven low-risk security threats. Acceptability was measured using the ISO/IEC 25010 standards, focusing on functional suitability, performance efficiency, and reliability. The results indicated that the system was "Acceptable" across all technical dimensions, regardless of the user's gender or employment status. Statistical analysis showed no significant difference in how different respondent groups perceived the system's usability. Ultimately, the study successfully proved that local car rental services can be enhanced through a secure, web-based platform that replaces traditional manual methods. The DMMMSU study shares a foundational goal with InstaCar in its mission to replace manual bookkeeping with a structured, digitalized rental ecosystem. Both systems are developed using web technologies like PHP and MySQL, which the 2023 study validates as reliable tools for local implementations. However, while the DMMMSU project focuses on a broad rental model, InstaCar introduces a more specialized "Tie-up" model that specifically empowers private car owners to act as providers. InstaCar also advances the rental logic by introducing distinct 10-hour and 24-hour booking tiers, whereas the DMMMSU system utilized a more traditional daily duration. The local study's findings on the "youth-heavy" demographic provide empirical support for InstaCar's choice of a mobile-responsive UI design to cater to tech-savvy Filipinos. Furthermore, while Abat et al. used the T-Test for acceptability, InstaCar can build upon this by using the Technology Acceptance Model (TAM) to measure deeper user intent. The vulnerability testing performed in the DMMMSU study serves as a security benchmark that InstaCar aims to exceed by implementing more robust database encryption. InstaCar also differs by focusing on a "shared economy" approach, whereas the local study acted as a centralized agency system. The 2023 research confirms that the local market accepts digital transitions, giving InstaCar a proven psychological "green light" for adoption in the Philippines. By citing this study, InstaCar

effectively bridges the gap between general academic acceptability and specific, highperformance fleet management. Finally, while the DMMMSU study highlights the needs of the La Union community, InstaCar is designed to be a scalable solution capable of adapting to various urban and provincial settings.

2.4 Related Literature

2.4.1 Evolution of Management Information Systems (MIS) and Sharing-Economy Frameworks

At its core, a Management Information System (MIS) represents a formal, sociotechnical, and organizational framework engineered to systematically collect, process, store, elevate, and distribute operational data across enterprise environments. Within modern commercial landscapes, information systems serve as the indispensable digital backbone for the sharing economy, facilitating the peer-topeer monetization of underutilized physical assets through centralized software interfaces. Academic literature on sharing economy frameworks consistently emphasizes that operational transparency and systemic trust serve as the primary drivers of user adoption across digital marketplaces. Information systems bridge this critical trust gap by maintaining verified user profiles, automating transparent pricing tiers, and generating immutable historical transaction logs for auditing purposes.

According to established software and management theories, the practical efficiency of a rental information system relies directly on its capacity to maintain real-time data accuracy across all system states. These platforms actively mitigate "information asymmetry"—an operational condition where one party possesses critical transactional data hidden from another—by making precise vehicle details, dynamic pricing structures, and real-time inventory schedules fully visible to both clients and administrators. Within the broader transportation logistics sector, information systems have evolved from basic database lookup tables into sophisticated multi-stakeholder ecosystem managers capable of balancing the interests of administrators, third-party fleet operators, and end consumers simultaneously.

2.4.2 Automated Financial Technology, Split Payments, and Verification Workflows

The integration of automated financial technology and compliance management modules has become a foundational pillar in reducing operational friction and securing digital marketplaces. Traditional car-sharing operations frequently suffer from severe booking drop-offs due to manual payment processing, off-platform banking transfers, and delayed security deposit returns. Contemporary literature in digital commerce highlights that embedding unified payment gateways allows for instant, end-to-end transaction handling via secure tokenization, which shields sensitive credit card credentials from the application’s primary database.

Beyond simple transactional clearing, these advanced financial layers automate split-payment mechanisms—a feature critical for consignment "Tie-up" marketplaces where the system must instantly calculate corporate commission fees, deduct platform insurance margins, and route remaining revenues directly to private vehicle owners. This automated accounting loop is systematically paired with digital identity verification workflows that validate driver's licenses and government credentials in real time. By automating financial settlement and security clearance, information systems eliminate the heavy human labour requirements of traditional checkout lines, effectively lowering overhead expenses for platform operators while fostering deep institutional trust among participants.

2.4.3 Data Privacy Laws, Cryptographic Security, and Role-Based Access Control (RBAC)

Underpinning interactive frontend environments is the critical requirement for robust, end-to-end data encryption and strict compliance with international privacy mandates such as the General Data Protection Regulation (GDPR), the California Consumer Privacy Act (CCPA), and local data privacy statutes (e.g., the Philippine Data Privacy Act of 2012). Because peer-to-peer and rental marketplaces collect exceptionally sensitive user payloads—including scanned driver's licenses, real-time location data, home addresses, and financial credit credentials—they represent high-value targets for malicious cyber-attacks. Security engineering literature notes that modern applications must enforce Transport Layer Security (TLS/HTTPS) across all active API endpoints to prevent packet-sniffing or maninthe-middle attacks during the data transmission phase.

At the data storage tier, private identifiers must be encrypted at rest using industry-standard algorithms such as AES256, while password strings are transformed via cryptographic hashing functions like bcrypt or Argon2 before hitting physical database disks. Furthermore, system administrators implement Role-Based Access Control (RBAC) configurations—a system architecture principle that grants user access strictly based on assigned roles within an organization. Security guidelines emphasize that restricting access rights (e.g., blocking regular users from administrative settings) protects sensitive database records, prevents unauthorized data changes, and ensures data integrity.

2.5 Synthesis of the Reviewed Literature

A cross-examination of contemporary international assessments and localized empirical studies reveals that traditional, analog car rental methods are fundamentally unequipped for a modern, data-driven economy—often leading to severe operational friction, double-bookings, financial leakage, and eroded customer trust as fleet complexity scales. Researchers consistently validate that transitioning to an automated, web-based platform built on robust database architectures, role-based access control (RBAC), and responsive interfaces is a strategic necessity to eliminate human error, streamline verification, and accelerate booking turnarounds. The **InstaCar** platform operationalizes these academic, technical, and socio-economic findings by replacing fragile manual ledgers with a comprehensive digital ecosystem that features an automated 10-hour and 24-hour pricing logic engine, a real-time dispatch calendar to eliminate overlapping reservations, an administrative analytics dashboard for data-driven decision-making, and a secure document verification queue to fulfil local data privacy and security mandates. By coupling these frontend usability and backend algorithmic controls with an inclusive "tie-up" consignment model, the platform addresses a critical market gap for flexible short-term mobility in the Philippines, providing a scalable, resilient, and community-driven framework that empowers independent vehicle operators, protects user privacy, and elevates regional transport logistics to modern enterprise standards.

2.3 Research Gap

While enterprise fleet management platforms offer comprehensive tracking features, their high licensing costs and rigid architectures render them impractical for independent local operators. Conversely, basic open-source web systems are lightweight but lack specialized micro-mobility pricing logic, automated partner profit-sharing, and realtime double-booking prevention algorithms. Furthermore, existing local rental workflows rely on manual paper logs or unencrypted messaging apps, creating severe data privacy vulnerabilities and operational bottlenecks. A clear research gap exists for an accessible, low-overhead, and secure web platform that unifies flexible short-term duration tiers (10-hour and 24-hour), automated consignment payouts, secure identity verification, and real-time dispatch synchronization tailored specifically to small and medium-sized rental enterprises.

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CHAPTER III

METHODOLOGY

3.1 System Architecture

The InstaCar platform employs a three-tier architecture to isolate interface design, core business logic, and data storage into distinct, independent operational layers. This separation improves system performance, simplifies maintenance, and enables concurrent Agile development while optimizing network throughput during peak user sessions. By preventing direct public access to the primary database and routing all traffic through server-side validation, the architecture strengthens data security and uniformly enforces key business rules, such as Role-Based Access Control (RBAC), flexible pricing policies, and real-time booking locks. Furthermore, this multi-layered framework simplifies debugging, reduces data latency, and provides a scalable, enterprise-ready foundation that easily supports future API integrations or multi-server hosting environments.

- **Client Side (Presentation Layer)**: Serves as the interactive web interface optimized for mobile and desktop viewports using a modern "Yellow, Black, and White" design theme. It features dynamic, role-based dashboards for Administrators, "Tie-up" vehicle owners, and Renters—enabling seamless inventory management, micro-tier bookings, and ID document uploads. Utilizing asynchronous requests for real-time updates and localized input validation, this tier catches entry errors instantly, saving network bandwidth and lowering server load.

- **Server Side (Application Layer)**: Built on PHP, this middle tier acts as the system's core intelligence and governance hub. It processes incoming requests, enforces Role-Based Access Control (RBAC), and manages user session security. This layer executes the core automated scheduling logic—preventing doublebookings—and securely handles mathematical computations (e.g., 5% down payments, PHP 200 carwash fees, and late penalties) to eliminate client-side manipulation.

- **Database (Data Layer)**: Utilizes a normalized MySQL relational database (RDBMS) to function as the permanent, secure ledger of the platform. By storing hashed user credentials, vehicle specs, document indexes, and transaction histories, it uses primary/foreign key constraints and atomic table locking to ensure data integrity during simultaneous bookings. Operating strictly behind the application layer, this tier prevents direct public access and utilizes optimized indexing for sub-second query performance.

3.2 Hardware and Software Environment

Table 4.1 presents the recommended hardware specifications.

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD
Monitor	1366 × 768 Resolution	Full HD Display
Internet Connection	10 Mbps	50 Mbps
Mobile Device	Android 10	Android 13 or higher

The recommended hardware configuration provides sufficient processing capability for executing analytical operations and rendering graphical dashboards without significant delays.

Table 4.2 presents the software requirements.

Windows 11 / Ubuntu Linux	Operating System
Visual Studio Code	

MySQL	Source Code Editor
	Database Management System
PHP / Node.js	Backend Development
ReactJS	Frontend Development
Google Chrome	Web Browser
Git	Version Control
XAMPP	Local Development Server

The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web application development.

3.3 Data Gathering Techniques

To ensure InstaCar addresses real-world operational bottlenecks rather than presumed problems, the researchers utilized a hybrid empirical data-gathering approach combining semi-structured interviews and document analysis across local transport operators in Iloilo Province. This field-driven data collection provided the core specifications for the platform's backend logic, relational database schemas, role-based dashboards, and security requirements.

- Interviews: Conducted with InstaCar's owner, these qualitative discussions provided first-hand insights into current manual workflows, cash-handling practices, identity verification friction, and owner-partner ("tieup") dynamics. The findings confirmed severe market frustration with double-bookings and manual fee calculations, validating the market demand for a decentralized owner dashboard and a flexible 10-hour rental duration tier alongside standard daily packages.

- Document Analysis: An exhaustive evaluation of physical paper logbooks, booking slips, rental contracts, and liability waivers was performed to map operational workflows directly into software rules. This analysis directly informed the system's automated billing logic—confirming the mandatory 5% reservation down payment, the PHP 200 carwash maintenance fee, and late-return penalty calculations. It also established key geographic policy boundaries, such as enforcing strict Panay Island travel limits within the digital terms of service.

By synthesizing field interview insights with physical document parameters, the researchers eliminated guesswork from the development pipeline. This triangulation ensured that every database column, algorithmic scheduling gate, and dashboard layout corresponds directly to verified, local commercial necessities.

To evaluate detection capabilities against unknown zero-day attacks, experimental traffic is generated using a hybrid benchmark dataset model:

3.4 Testing Procedures

To guarantee structural integrity and prevent database corruption or logical errors during live transaction flows, InstaCar undergoes a progressive, three-tier testing framework. This structured methodology moves from isolated backend code modules to complete system-wide integrations and end-user field evaluations:

- **Unit Testing**: Focuses on evaluating individual source code functions, input forms, and database scripts in isolation within a local development environment. It validates critical boundary rules—such as input field formatting (e.g., license plates and pricing), password hashing, session management cleanup, and sanitization routines—to ensure invalid data or malicious code (SQLi/XSS) are intercepted before system integration.

- **System Testing**: Evaluates the entire platform as a unified, fully integrated ecosystem across the presentation layer, PHP application engine, and MySQL database. This tier tests end-to-end booking workflows, verifying that selected duration tiers (10-hour vs. 24-hour) correctly calculate the 5% reservation down payment, the PHP 200 carwash fee, and late return penalties. It also validates database relational constraints and atomic table locking to prevent concurrent double-booking conflicts and enforce Role-Based Access Control (RBAC) boundaries.

- **User Acceptance Testing (UAT)**: Represents the final evaluation milestone, engaging target end-users—including local commuters, independent "tie-up" vehicle owners, and IT evaluators—to measure real-world usability. Evaluators interact with the live prototype, navigating the "Yellow, Black, and White" interface, uploading sample identification documents, and executing mock reservations. Standardized usability surveys and feedback forms gather quantitative metrics to certify that navigation paths, pricing breakdowns, and fleet management tools accommodate the technical literacy of local users.

3.5 System Development Methodology

The researchers adopted the **Agile Development Methodology** as the operational framework for engineering **InstaCar**. Agile's flexible, collaborative, and iterative sprint structure enabled the team to break down the platform's complex architecture into manageable modules, facilitating early detection of logical bugs, continuous refinement of user requirements, and dynamic prioritization of high-risk components (such as the double-booking conflict algorithm).

The engineering process executed across five distinct Agile phases:

- **Requirements Analysis:** Gathered operational needs through interviews and fieldwork observation to define a clear project backlog. Mapped four core vehicle availability states (*Available, Not Available, Rented, Under Maintenance*) and established computational rules for **10-hour vs. 24-hour duration tiers**, mandatory **5% reservation down payments**, fixed **PHP 200 carwash fees**, and late return penalties. Nonfunctional specifications (security, privacy, and sub-second query latency) were structured into actionable user stories.
- **System Design:** Formulated UI/UX wireframes and database schemas to translate requirements into technical blueprints. Defined a mobile-responsive "Yellow, Black, and White" interface across dedicated dashboards for Administrators, "Tie-up" vehicle owners, and Renters. Database tables were normalized using Entity-Relationship Diagrams (ERD) and Data Flow Diagrams (DFD) to enforce Role-Based Access Control (RBAC) and eliminate data redundancy.

- **Development:** Built the functional, data-driven web application using **PHP** for server-side business logic and session management, and **MySQL** (via XAMPP) for relational data storage. Programmed the core scheduling algorithms, secure password hashing, dynamic invoice computation with decimal precision, and encrypted driver's license document upload directories under version control.
- **Testing:** Executed rigorous diagnostic cycles—moving from isolated **Unit Testing** (input form validation, SQLi/XSS sanitization) to full **System Testing** (end-to-end booking transactions, atomic schedule locking, penalty logic) and **User Acceptance Testing (UAT)**—ensuring mathematical precision, cross-browser compatibility, and fault-tolerant handling of edge cases.
- **Deployment:** Staged the final, optimized software prototype for local hosting and defense demonstration. Compiled detailed user manuals, fine-tuned server configuration paths, established automated database backup snapshots, and established post-deployment feedback loops for future cloud migration.

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CHAPTER IV

RESULTS AND DISCUSSION

4.1 Overview of the Developed System

The Car Rental Management and Analytics System was developed to modernize car rental operations by replacing inefficient manual methods—such as paper documents and spreadsheets—with a centralized, automated digital platform. Manual processes previously caused operational friction, including delayed transactions, duplicate records, and difficulties in tracking vehicle availability and rental histories. To resolve these challenges, the system automates core operations, allowing administrators and staff to manage customer records, vehicle information, reservations, payments, and maintenance schedules efficiently with minimal training. Built with a simple and user-friendly interface,

the platform utilizes a centralized database to improve record accuracy, reduce manual workload, and ensure easy information retrieval. Beyond administrative automation, the system features an analytics dashboard that generates reports and graphical summaries on rental trends, vehicle utilization, revenue, and key performance indicators, providing business owners with valuable data-driven insights to enhance daily monitoring, planning, and decision-making.

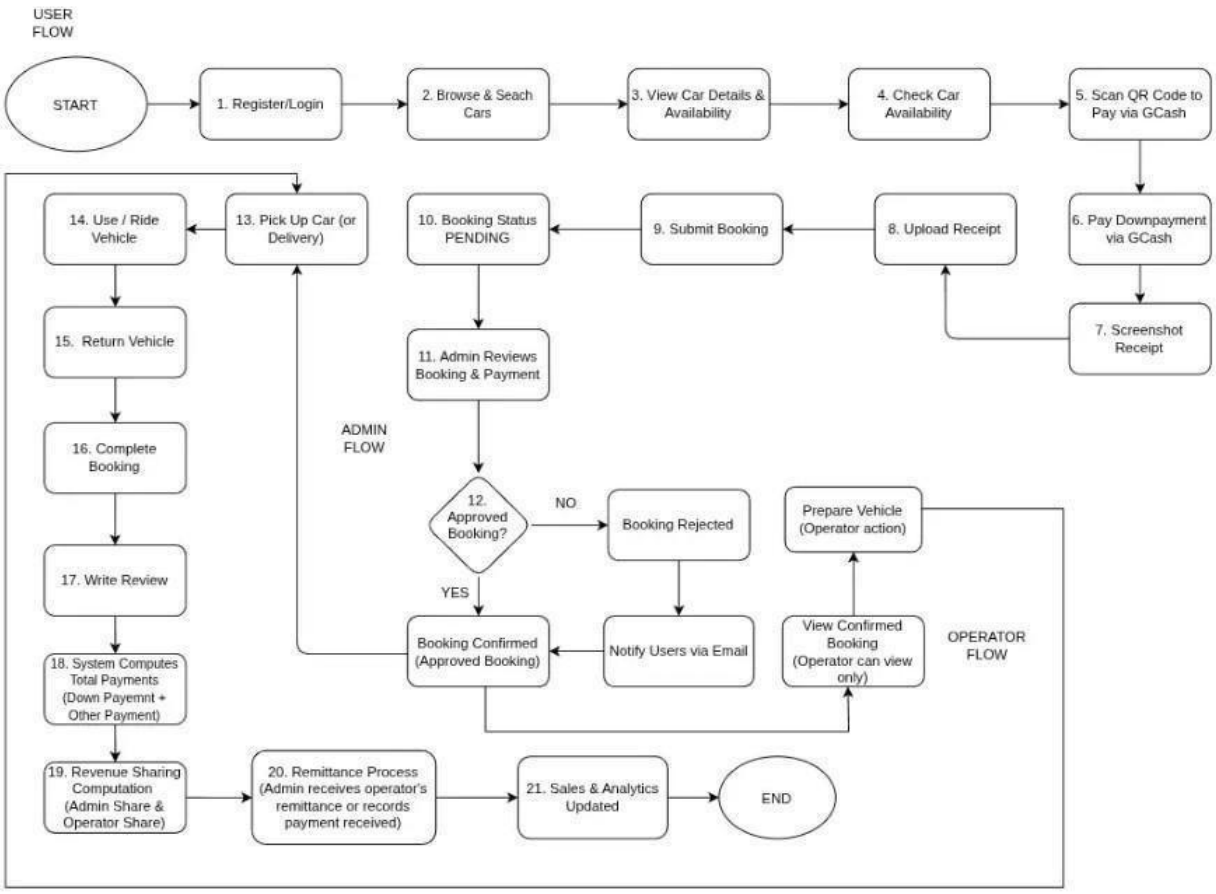
The developed system was created with the following objectives:

- To automate the management of customer, vehicle, and rental records.
- To simplify the reservation and rental transaction process.
- To monitor the availability and current status of rental vehicles.
- To record customer payments and maintain rental histories accurately.
- To generate reports on rentals, revenue, and vehicle utilization.
- To provide analytical dashboards that supported business monitoring and decisionmaking.
- To improve the overall efficiency, accuracy, and security of car rental operations.

The development of the **Car Rental Management and Analytics System** followed modern software development practices. It incorporated a responsive user interface, secure login and authentication, centralized database management, and reporting tools that made daily operations more organized and efficient. These features ensured that only authorized users could access the system while maintaining the accuracy and security of stored information.

The completed system consisted of several integrated modules that worked together to support the different functions of a car rental business. These modules included user authentication, customer management, vehicle management, reservation processing, rental management, payment management, maintenance monitoring, and analytics reporting. Since all modules shared the same database, information was updated automatically, allowing users to view accurate and real-time records throughout the system.

Figure 4.1 presents the overall workflow of the developed system.

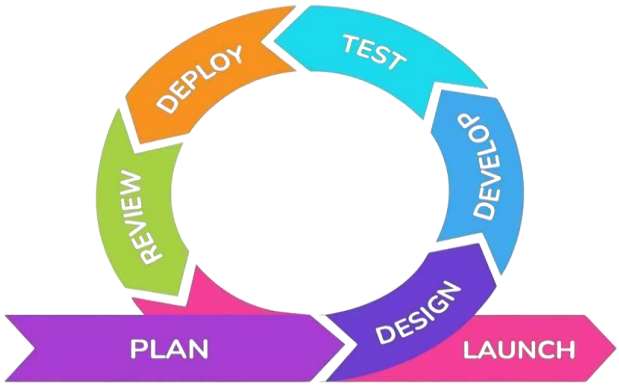


The booking workflow began with user authentication through account creation or login, enabling the customer to browse vehicles, review specifications, rates, and real-time availability. Upon selecting a car, the user submitted a booking request by scanning a GCash QR code to pay the required down payment and uploading a transaction screenshot, which set the reservation status to **Pending**. The administrator then verified the booking and payment details, updating valid requests to **Booking Confirmed** or issuing an automated rejection email if discrepancies were found. Following approval, the operator prepared the vehicle for customer pickup or delivery for the duration of the rental period. Upon the vehicle's return, the system updated the status to **Completed** and allowed the user to submit ratings and feedback. Automatically, the system computed the total payment, settled remaining balances, calculated the profit-sharing distribution between the administrator and operator, and logged the remittance records. Finally, all completed transactions were securely archived in the centralized database and routed to the **Sales and Analytics** module, updating real-time reports and summaries to support administrative monitoring and data-driven decision-making.

4.2 System Development Methodology

The development of the **Car Rental Management and Analytics System** followed the **Agile Software Development Methodology**, utilizing an iterative and collaborative approach to ensure continuous feedback, incremental improvement, and dynamic adaptation to operational needs across seven distinct phases:

- **Requirements Gathering:** Empirical data was collected through client interviews, workflow observations, and document analyses to identify key operational challenges—such as manual tracking errors, delayed reporting, and a lack of business analytics—establishing the core functional and non-functional system parameters.
- **System Planning:** Defined the project scope, timeline, hardware/software resource requirements, core operational modules, and conducted a comprehensive feasibility analysis (technical, operational, and economic) to ensure structured project execution.
- **System Design:** Formulated application blueprints using Unified Modeling Language (UML) diagrams, Entity-Relationship Diagrams (ERDs), Data Flow Diagrams (DFDs), flowcharts, and wireframes to establish an intuitive, mobile-responsive user interface and a normalized, consistent database structure.
- **Software Development:** Translated design specifications into functional code module-by-module. Programmed core functionalities—including secure authentication, customer/vehicle management, reservation processing, payment recording, maintenance tracking, and an analytics dashboard for graphical business intelligence—using incremental integration to correct errors early.
- **Testing:** Executed a multi-tiered evaluation pipeline comprising **Unit Testing** for isolated functions, **Integration Testing** for inter-module data flow, **System Testing** for overall performance and security, and **User Acceptance Testing (UAT)** evaluated against the **ISO/IEC 25010 Software Quality Model** to refine usability based on end-user feedback.
- **Deployment:** Installed and configured the finalized application and database in the operational environment. Administered system orientation, set up role-based user accounts, and validated live transaction workflows with sample operational data to verify system stability.
- **Maintenance:** Implemented post-deployment performance monitoring, minor software updates, and security optimizations, while documenting user feedback for potential future enhancements such as online payment gateway integrations, automated SMS/email alerts, and mobile application expansion.



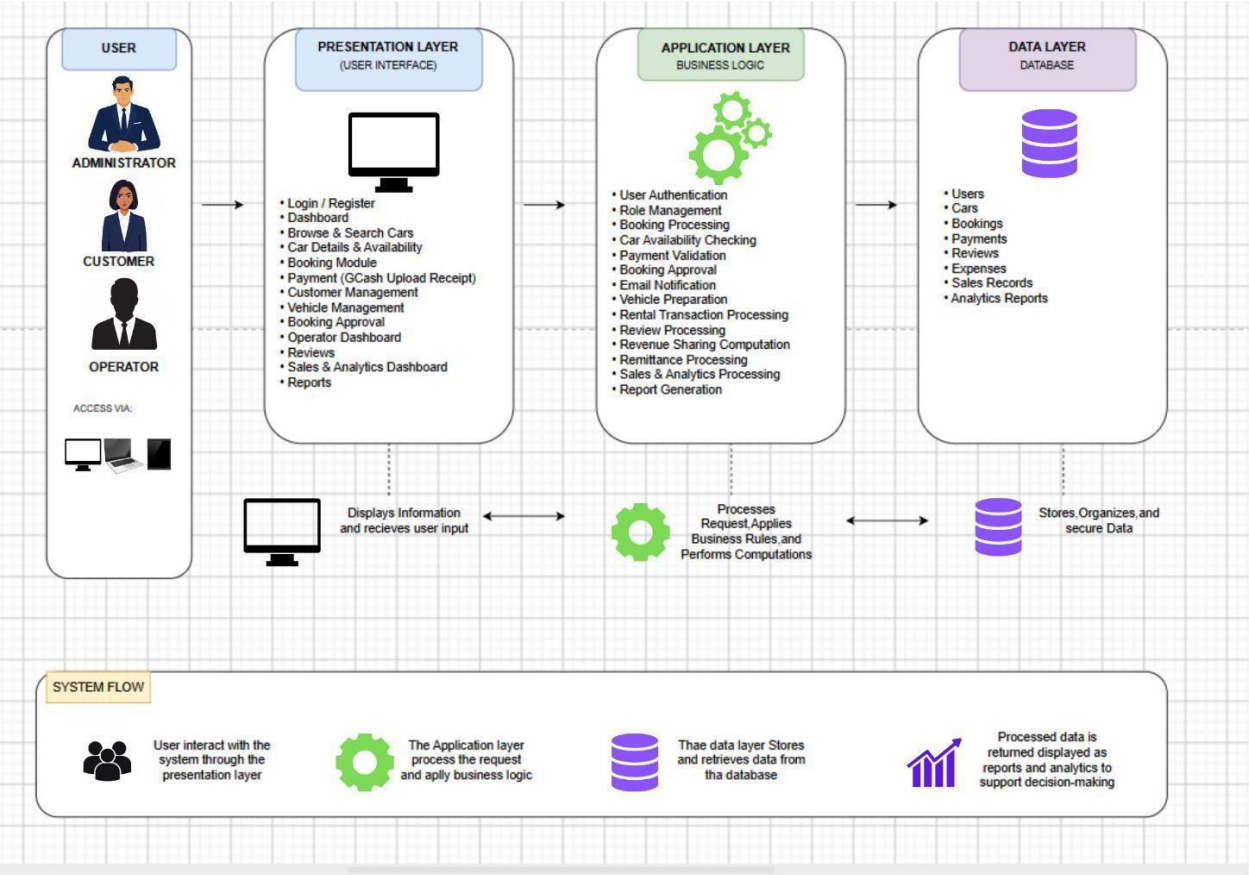
4.3 System Architecture

The **Car Rental Management and Analytics System** utilized a **Three-Tier Client-server**

Architecture, which divided the system into three independent but interconnected layers: the **Presentation Layer**, **Application Layer**, and **Data Layer**. This architectural design was selected because it improved the organization of the system by separating the user interface, business logic, and database management into different layers. As a result, the system became easier to maintain, more secure, and capable of handling future enhancements without significantly affecting the existing functionalities.

The Three-Tier Architecture also allowed the application to process user requests more efficiently by assigning specific responsibilities to each layer. This separation reduced system complexity, improved maintainability, and ensured that data could be securely managed while providing users with a smooth and responsive experience. The architecture supported the overall goal of developing a reliable and scalable system that could efficiently manage car rental operations and generate analytical reports for business decision-making.

Figure 4.3 illustrates the overall system architecture of the developed **Car Rental Management and Analytics System**.



The **Car Rental Management and Analytics System** is built on a **Three-Tier Client-Server Architecture** that isolates the user interface, core business logic, and database management into distinct, interconnected operational layers:

1. Presentation Layer (Front-End Interface)

- **Role:** Acts as the primary user-facing layer accessible across smartphones, tablets, laptops, and desktop browsers via internet or local network connection.

- **Key Components:** Features responsive, easy-to-navigate modules including the **Login Page**, **Dashboard**, **Customer**
  Management, **Vehicle Management**, **Reservation Management**, **Rental Management**,
  Payment Management, **Maintenance Management**, **Analytics Dashboard**, and printable **Reports**.

2. Application Layer (Core Business Logic)

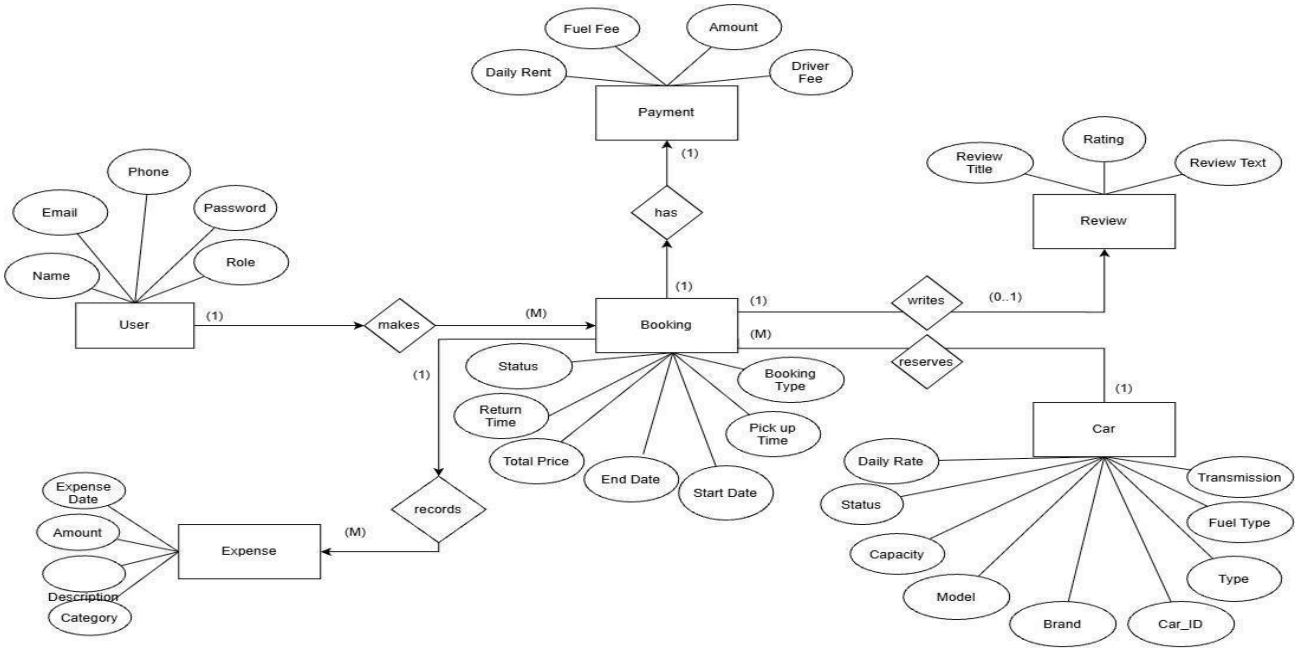
- **Role:** Functions as the communication bridge between the user interface and database, enforcing business rules, data validation, role-based access control, and user authentication.
- **Core Operations:** Manages all transaction execution, input verification, revenue computations, rental trend tracking, vehicle utilization analysis, and dynamic report/dashboard generation before passing or retrieving data from storage.

3. Data Layer (Central Repository)

- **Role:** Utilizes a normalized Relational Database Management System (RDBMS) using primary and foreign keys to store, organize, and retrieve operational and analytical data while preventing data redundancy and enforcing access security.
- **Stored Datasets:** Houses user accounts, customer information, vehicle records, categories, reservations, rental transactions, payments, maintenance logs, availability statuses, revenue logs, analytics data, and historical reports.
- **Data Integration:** Instantly synchronizes actions across modules—such as automatically updating fleet availability, logging payments, and refreshing the analytics dashboard upon transaction completion.

4.4 Database Design

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



The database of the Car Rental Management and Analytics System utilizes a centralized relational model to store user accounts, vehicle records, bookings, payments, reviews, and expenses. By implementing primary and foreign keys across structured tables, it minimizes redundancy, maintains referential integrity, and ensures data

consistency. This organized design enables efficient data processing and seamless synchronization across modules to support real-time administrative reporting and analytics. The primary entities of the database included the following:

- Users
- Cars
- Bookings
- Payments
- Reviews
- Expenses

Users Table

The **Users** table stored the account credentials and profile information of authorized users who could access the system. It was responsible for authentication and determining the access privileges of each user.

Field	Description
user_id	Primary Key
name	Full Name of the User
email	User Email Address
phone	Contact Number
password	Encrypted Password
role	User Role (Administrator or Staff)

Cars Table

The Cars table stored detailed information about every vehicle available for rent. It is also maintained the current status of each vehicle to help administrators monitor availability.

Field	Description
car_id	Primary Key
brand	Vehicle Brand
model	Vehicle Model
type	Vehicle Type
transmission	Transmission Type
fuel_type	Fuel Type
capacity	Seating Capacity
daily_rate	Daily Rental Rate
status	Vehicle Status

Bookings Table

The **Bookings** table recorded all reservation and rental transactions made by customers. It served as the central table of the system because it connected users, vehicles, payments, and reviews.

Field	Description
booking_id	Primary Key
user_id	Foreign Key
car_id	Foreign Key
start_date	Rental Start Date
end_date	Rental End Date
pickup_time	Vehicle Pick-up Time
return_time	Vehicle Return Time
booking_type	Booking Type
total_price	Total Rental Cost
status	Booking Status

Payments Table

The **Payments** table stored all payment information associated with rental transactions. It recorded the different charges incurred during the rental period and the total amount paid by the customer.

Field	Description
payment_id	Primary Key
booking_id	Foreign Key
daily_rent	Daily Rental Fee
driver_fee	Driver Service Fee
fuel_fee	Fuel Charge
amount	Total Payment Amount

Reviews Table

The **Reviews** table stored customer feedback submitted after completing a rental transaction. These records allowed administrators to evaluate customer satisfaction and improve service quality.

Field	Description
review_id	Primary Key
booking_id	Foreign Key
review_title	Review Title
review_text	Customer Feedback
rating	Customer Rating

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Expenses Table

The **Expenses** table recorded the operational expenses of the car rental business. It helped administrators monitor business expenditures and maintain financial records.

Field	Description
expense_id	Primary Key
category	Expense Category
description	Expense Description
amount	Expense Amount
expense_date	Date of Expense

The relationships established among these tables enabled the system to efficiently manage rental operations while maintaining accurate and consistent records. The **Users** table was linked to the **Bookings** table, allowing the system to identify the user responsible for each booking transaction. Each booking was associated with a specific vehicle through the **Cars** table and was connected to its corresponding **Payment** record to ensure accurate tracking of rental charges. Completed bookings could also be linked to customer **Reviews**, providing valuable feedback regarding the quality of service. In addition, the **Expenses** table-maintained records of operational costs, allowing administrators to monitor business expenditures alongside rental income.

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

The development and evaluation of the Car Rental Management and Analytics System demonstrated that the objectives of the study were successfully achieved. The findings showed that the developed system effectively addressed the challenges associated with the manual management of car rental operations by providing a centralized and automated platform for managing business transactions.

Based on the results of the study, the following findings were obtained:

- The developed system successfully automated customer records, vehicle management, reservations, payments, and maintenance monitoring, resulting in faster and more organized operations.
- The system effectively monitored vehicle availability and status, enabling easy identification of available, rented, or maintenance units to improve tracking accuracy and eliminate scheduling conflicts.
- The analytics dashboard transformed transaction data into actionable reports and graphical summaries on trends, vehicle utilization, and revenue to support data-driven decision-making.
- The centralized database enhanced data accuracy, consistency, and security by minimizing duplicate entries, reducing manual errors, and providing quick access for authorized users.
- The system generated comprehensive reports that simplified daily monitoring of rental activities, customer transactions, and overall business performance while reducing manual documentation time.
- Based on the system evaluation using the ISO/IEC 25010 Software Quality Model, respondents rated the application as functional, reliable, user-friendly, efficient, secure, maintainable, and portable.
- User Acceptance Testing (UAT) confirmed the system's ease of use, showing that respondents executed daily operational tasks with minimal difficulty.

Overall, the findings of the study showed that the Car Rental Management and Analytics System successfully improved operational efficiency by automating core processes and delivering analytical tools for informed decisionmaking, demonstrating its value for modern car rental management.

5.2 Conclusions

The Car Rental Management and Analytics System proved to be an effective and practical solution for modernizing daily car rental operations. By automating essential processes—including customer

management, vehicle tracking, reservations, payments, and reporting—the system eliminated the limitations of manual record-keeping and significantly improved operational efficiency.

The integration of analytics features successfully transformed transaction data into actionable reports and visual summaries. These insights into rental trends, vehicle utilization, and revenue performance empowered business owners to make informed, data-driven decisions and optimize resource management using organized, real-time information.

Overall, the study concluded that the system serves as a reliable solution that simplifies business workflows, enhances administrative productivity and record accuracy, and supports effective long-term business management through robust reporting and analytics capabilities.

5.3 Recommendations

Based on the findings and conclusions of the study, the researchers proposed the following recommendations to further improve the Car Rental Management and Analytics System and enhance its effectiveness for future implementation:

System Enhancements

- The system could be enhanced by integrating online payment gateways to allow customers to make secure and convenient online payments using digital wallets, bank transfers, or credit and debit cards.
- An SMS and email notification feature could be added to automatically remind customers about reservation confirmations, payment due dates, rental schedules, vehicle return dates, and maintenance updates. This feature would improve communication between the business and its customers.
- Future versions of the system could include a mobile application for Android and iOS devices to provide users with greater accessibility and allow administrators to monitor rental activities anytime and anywhere.
- The analytics dashboard could be expanded by incorporating advanced reporting and forecasting features, such as predicting peak rental seasons, identifying customer rental patterns, and estimating future business demand. These additional insights could further support strategic planning and business decisionmaking.
- A GPS vehicle tracking feature could also be integrated to allow administrators to monitor the realtime location of rented vehicles, improving vehicle security and operational efficiency.

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